

Package ‘HZIP’

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Type Package

Title Likelihood-based inference for joint modeling of correlated count and binary outcomes with extra variability and zeros

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Description Likelihood-based inference for joint modeling of correlated count and binary outcomes with extra variability and zeros.

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URL <https://github.com/carrascojalmar/HZIP>

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envelope.HZIP	<i>Envelope simulation for HZIP Model</i>
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Description

Produces a Q-Q plot of residuals from a Hierarchical Zero-Inflated Poisson (HZIP) Model fitted via [hzip](#).

Usage

```
envelope.HZIP(object, nsim = 100, ...)
```

Arguments

- object An object of class HZIP, typically returned by [hzip](#).
- nsim Integer. Number of simulations used to construct the envelope. Default is 100.
- ... Additional arguments (currently ignored).

Details

A simulation envelope is added using Monte Carlo replications.

Value

Envelope simulation plot.

See Also

[hzip](#), [residuals.HZIP](#)

Examples

```
## Not run:
data(salamanders)
fit.salamander <- hzip(y ~ mined|mined+spp,data = salamanders)
res <- residuals(fit.salamander)
envelope.HZIP(res, nsim = 5000)

## End(Not run)
```

hzip

Fit a Hierarchical Zero-Inflated Poisson (HZIP) Model

Description

`hzip()` fits a longitudinal/clustered zero-inflated Poisson model with subject-level random effects by maximizing a (marginal) likelihood approximated. The model uses a two-part [Formula](#): y zero part | count part, where the count intensity (Poisson mean) and the zero-inflation probability are linked to (possibly different) sets of covariates. Initial values are obtained from `pscl::zeroinfl(..., dist = "poisson", link = "cloglog")`.

Usage

```
hzip(
  formula,
  data,
  hessian = TRUE,
  method = "BFGS",
  Q = 15,
  lower = -Inf,
  upper = Inf,
  control = NULL,
  ...
)
```

Arguments

formula	A two-part Formula of the form $y \sim w_zero + \dots \mid x_count + \dots$, where the right-hand side before the bar specifies covariates for the zero-inflation component and the right-hand side after the bar specifies covariates for the Poisson mean.
data	A <code>data.frame</code> containing all variables used in <code>formula</code> and a subject identifier named <code>Ind</code> (one row per observation).
hessian	Logical; if <code>TRUE</code> (default) the observed Hessian at the optimum is returned and used to compute standard-error estimates.
method	Character string passed to optim (default "BFGS").
Q	Integer; number of Gauss–Hermite nodes for quadrature (default 15). Larger values improve accuracy at higher computational cost.
lower	Bounds on the variables for the "L-BFGS-B" method, or bounds in which to search for method "Brent" (arguments passed to optim).
upper	method, or bounds in which to search for method "Brent" (arguments passed to optim).
control	Optional list passed to optim 's <code>control</code> = argument (e.g., <code>list(maxit = 500)</code>).
...	Further arguments passed to optim .

Details

Let y_{ij} denote the count response for subject i at occasion j . The HZIP model assumes

$$P(y_{ij} = 0 \mid u_i) = \pi_{ij}(u_i) + \{1 - \pi_{ij}(u_i)\} \exp\{-\mu_{ij}(u_i)\},$$

$$P(y_{ij} = k \mid u_i) = \{1 - \pi_{ij}(u_i)\} \frac{\mu_{ij}(u_i)^k e^{-\mu_{ij}(u_i)}}{k!}, \quad k \geq 1,$$

with linear predictors for the count and zero parts (links typically log for the Poisson mean and cloglog for the zero-inflation). Subject-specific random effects u_i induce within-subject dependence; the marginal likelihood is approximated by Gauss–Hermite quadrature with Q nodes.

Value

An object of class "HZIP", a list with elements:

call	The matched call.
formula	The model Formula.
coefficients_zero	Estimated coefficients for the zero-inflation part.
coefficients_count	Estimated coefficients for the count part.
scale_zero	Estimated scale (zero part).
scale_count	Estimated scale (count part).
loglik	Optimized objective value returned by optim. (Note: depending on lvero, this may be the negative log-likelihood.)
convergence	optim convergence code.
n	Number of observations or subjects (see Note).
m	Cluster sizes per subject (vector ordered by Ind).
ep	Approximate standard errors (square roots of the diagonal of the inverse Hessian).
iter	Number of optim iterations.
method	Optimization method.
optim	Raw optim output.
data	The input data.

Note

The subject identifier must be named Ind. The sign convention for the zero-part coefficients in the initial values follows `pscl::zeroinfl`; the internal parameter vector is `c(scale_zero, -beta_zero, scale_count, beta_count)`. Also verify whether `loglik` is the (negative) log-likelihood as returned by your objective `lvero`; if it is the negative log-likelihood, you may want to store `logLik = -op$value` for user convenience.

References

- Min, Y., & Agresti, A. (2005). Random effect models for repeated measures of zero-inflated count data. *Statistical Modelling*, 5(1), 1–19.
- Jackman, S. (2020). *pscl: Classes and Methods for R Developed in the Political Science Computational Laboratory*. R package version 1.5.5.
- Zeileis, A., & Croissant, Y. (2010). Extended model formulas in R: *Journal of Statistical Software*, 34(1), 1–13. (**Formula**)

Examples

```
## Not run:
fit.salamander <- hzip(y ~ mined|mined+spp,data = salamanders)
summary(fit.salamander)

## End(Not run)
```

residuals.HZIP

*Compute Residuals for HZIP Models***Description**

This function calculates residuals for objects of class HZIP using randomized quantile residuals. The computation is performed efficiently using C++ functions for predicting random effects and calculating residuals.

Usage

```
residuals.HZIP(object, ...)
```

Arguments

object	An object of class HZIP, typically returned from hzip .
...	Additional arguments (not used).

Details

The function internally groups the data by individual (Ind), constructs model matrices for both zero-inflation and count parts of the model, and then calls the C++ functions `predict_HZIP_cpp_vec` and `r_ij_cpp_vec` to efficiently compute the residuals. Random effects are integrated using adaptive quadrature based on the supplied nodes and weights.

Value

A numeric vector of residuals with length equal to the total number of observations in the dataset.

Examples

```
## Not run:
fit.salamander <- hzip(y ~ mined|mined+spp,data = salamanders)
residuals(fit.salamander)

## End(Not run)
```

rHZIP	<i>Simulate Data from a Hierarchical Zero-Inflated Poisson (HZIP) Model</i>
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Description

rHZIP() generates panel/longitudinal data from a two-part hierarchical zero-inflated Poisson model with subject-specific random effects for both the zero-inflation and the count components. Random effects are drawn from a generalized log-gamma (GLG) distribution via rgengamma() (user must provide/attach a function with this name; see **Dependencies**).

Usage

```
rHZIP(n, m, para, x1, x2)
```

Arguments

- | | |
|------|---|
| n | Integer. Number of subjects. |
| m | Integer vector of length n (or a scalar recycled to length n). Numbers of repeated measurements per subject. |
| para | Numeric vector of parameters in the order c(lambda1, beta1, lambda2, beta2), where: <ul style="list-style-type: none"> • lambda1: GLG scale/shape parameter for the zero part. • beta1: length-p1 vector of coefficients for the zero part (matches ncol(x1)). • lambda2: GLG scale/shape parameter for the count part. • beta2: length-p2 vector of coefficients for the count part (matches ncol(x2)). Internally, the linear predictors are $\eta_{ij}^{(0)} = x_{1,ij}^\top \beta_1 + b_i^{(0)}$ and $\eta_{ij}^{(1)} = x_{2,ij}^\top \beta_2 + b_i^{(1)}$, with $p_{ij} = 1 - \exp\{-\exp(\eta_{ij}^{(0)})\}$ (cloglog link for zero-inflation) and $\mu_{ij} = \exp(\eta_{ij}^{(1)})$ (log link for counts). |
| x1 | Numeric matrix of covariates for the zero-inflation part (dimension sum(m) by p1). Include an intercept column if desired. |
| x2 | Numeric matrix of covariates for the count part (dimension sum(m) by p2). Include an intercept column if desired. |

Details

For subject $i = 1, \dots, n$ with m_i observations, the model draws two subject-level random effects $b_i^{(0)}$ and $b_i^{(1)}$ independently from GLG distributions parameterized by lambda1 and lambda2. Conditional on these effects, outcomes are generated as

$$Y_{ij} = Z_{ij} \times C_{ij},$$

where $Z_{ij} \sim \text{Bernoulli}(p_{ij})$ controls structural zeros and $C_{ij} \sim \text{Poisson}(\mu_{ij})$ controls the count size.

The returned data frame contains subject IDs, the response y, and the supplied covariates. An attribute "propZeros" stores a small summary with the percentage of structural zeros, additional Poisson zeros, and total zeros.

Value

A tibble with columns:

Ind	Subject identifier (1..n), repeated according to m.
y	Simulated response.
x*, w*	The covariates from x1 and x2 (renamed as described below).

The object has an attribute "propZeros": a 3×1 data.frame with rows "Zeros" (structural zeros, %), "Count" (extra zeros from the Poisson part, %), and "Total" (overall zero percentage).

Column naming

Columns of x1 are renamed to x1, x2, ..., xp1. Columns of x2 are copied except the first column is dropped and the remaining are renamed w1, w2, ..., w_{p2-1}. (This mirrors the current implementation that excludes the first x2 column from the output.)

Dependencies

This function calls `rgengamma()` to draw GLG random effects. Ensure that such a function is available on the search path (e.g., from a package that provides a generalized log-gamma RNG) or provide your own implementation with the signature `rgengamma(n, mu, sigma, lambda)`. It also uses **dplyr** and **tibble**.

See Also

[hzip](#) for model fitting.

Examples

```
## Not run:
set.seed(123)

n <- 50
m <- rep(4, n)
N <- sum(m)

# design matrices (with intercepts)
x1 <- cbind(1, rnorm(N))
x2 <- cbind(1, rnorm(N), rbinom(N, 1, 0.5))

p1 <- ncol(x1); p2 <- ncol(x2)
lambda1 <- 0.7
beta1 <- c(-0.2, 0.6)
lambda2 <- 0.9
beta2 <- c(0.3, 0.5, -0.4)

para <- c(lambda1, beta1, lambda2, beta2)

sim <- rHZIP(n, m, para, x1, x2)
head(sim)
attr(sim, "propZeros")

## End(Not run)
```

`salamanders`*Salamanders data*

Description

This dataset is adapted from the **glmmTMB** package and contains salamander counts with information on mining status and species. It is intended for illustrating zero-inflated Poisson models with random effects using the `hzip()` function.

Usage

```
salamanders
```

Format

A data frame with 644 rows and 4 variables:

Ind Individual identifier (factor).

y Count response variable (integer).

mined Mining status: "yes" or "no".

spp Species factor with multiple levels (e.g., GP, PR, DM, etc.).

Details

The dataset was originally included in the **glmmTMB** package (Brooks et al., 2017), and has been slightly modified for testing the **HZIP** package.

Source

Adapted from the **glmmTMB** package.

Examples

```
data(salamanders, package = "HZIP")

## Fit zero-inflated Poisson with random effects
fit.salamander <- hzip(y ~ mined + spp | mined + spp + mined * spp,
  data = salamanders)
summary(fit.salamander)
```

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